

UNRESOLVED ATTRIBUTION IS NOT NEGATIVE EVIDENCE

A forensic case study of document-state transitions, authorization reversals, cloud synchronization, publication-state divergence, and AI application activity

Abstract

This paper develops a forensic reasoning framework for complex cloud and AI application environments in which material events can be established at one evidentiary layer while actor, intent, or end-to-end causation remains unresolved at another. The record contains same-object destructive document transitions, contradictory automatic authorization outcomes on unchanged returned content, successful file-specific OneDrive synchronization, remote Git object transfer without corresponding publication-branch advancement, controlled Windows shell-history reproduction, observed ChatGPT/Codex process lineage, and contemporaneous service-endpoint correlations. The central result is methodological: uncertainty at a higher attribution layer must not be propagated backward to erase an event already established by lower-level receipts.

1. Thesis and Evidence Model

The core analytical rule is simple: forensic uncertainty propagates forward into propositions that require missing evidence. It does not propagate backward into propositions already supported by direct receipts. The source record states the rule operationally: cause unresolved does not subtract from failure observed; actor unresolved does not subtract from destructive edit observed; payload unresolved does not subtract from network transfer observed. [Full.PDF, p. 64]

The evidence is best understood as a ladder: observation; exact object identity; state transition; process attribution; service attribution; application operation; account attribution; human attribution; and intent. A finding at a lower rung is not invalid because a higher rung remains unknown. Equally, lower-rung evidence cannot be promoted merely because surrounding chronology is suggestive.

2. Same-Object Destructive Document Transitions

Eight restored Google Drive objects had substantive earlier revisions followed by later empty revisions. Seven destructive transitions occurred within 245.622 seconds on September 7; an eighth followed on September 8. The recovered earlier texts total 74,609 normalized characters. [Full.PDF, p. 83]

The proposition is a state transition: substantive revision -> later empty revision. Revision metadata attributes later events to the signed-in account perspective but does not identify the physical person, browser session, application, connector, or automation. [Full.PDF, p. 83]

One Word-document branch is more direct: an older retained revision yielded 46,092 characters; a later 7,683-byte DOCX contained one empty body paragraph, zero text nodes, no media, and no embedded documents, followed by a Trash move roughly a minute later. This rules out mere extraction failure as the explanation for the blank state. [Full.PDF, p. 56]

Finding: destructive document-state transitions are established. Initiating mechanism and actor are separate attribution questions.

3. Automatic Authorization Reversal

The same Drive object, Coup3.txt, was successfully read, blocked by automatic approval review, then successfully read twice more. Each successful read returned 74,025 characters with the same SHA-256. No scope-expanding instruction was recorded between the rejection and subsequent successful reads. [Full.PDF, pp. 83-84]

The observable sequence is ALLOW -> BLOCK -> ALLOW -> ALLOW on unchanged successfully returned content. The rejection was semantic, not a timeout or missing-file error. [Full.PDF, p. 55]

This establishes inconsistent visible authorization behavior. An unexposed review-state variable may explain the reversal, but explanation and negation are different operations.

4. File-Specific OneDrive Synchronization

The OneDrive evidence crosses from network correlation to application-level transfer telemetry. The diagnostics contain 292 successful upload-completion records: 290 repeated updates of events.jsonl and two of access-events.jsonl. The records identify local file IDs or paths, successful completion state, request identifiers, and OneDrive/SharePoint endpoints. [Full.PDF, p. 85]

A later trace is stronger: OneDrive.exe repeatedly read the exact growing monitor file, totaling 2,327,958 reported read bytes, followed roughly 58-65 ms later by 24 TCP send events totaling 2,346,190 reported bytes. Read sizes and send totals grew by corresponding increments. [Full.PDF, p. 85]

The proposition is therefore named-file synchronization, not merely a suspicious socket. Later cloud-reader identity is a different question.

5. Git Object Transfer Versus Publication

Repository evidence shows why upload and publication must be treated as distinct states. Numerous successful Git tree creations existed while the publication branch at an earlier cutoff

was only one commit ahead with one added file rather than a completed 259-document archive. [Full.PDF, p. 84]

The later consolidated finding states that a verified remote object set contained 292 files, including 289 research-document editions, and that the completed tree existed remotely as a3c7d162d9133964204cb5ddcf740e962cb9a5eb while the publication branch did not advance to that tree. [Full.PDF, p. 101]

The precise finding is: the completed Git object set was transmitted into remote object storage, while branch publication did not advance to expose the completed state. Object transfer and ref advancement are different operations.

6. Controlled Shell Reproduction

A controlled experiment used a uniquely named .py file through the Windows Explorer ChatGPT OpenWith route. The test changed 4183059cb0582.automaticDestinations-ms and caused the unique filename to appear inside the modified Jump List container. [Full.PDF, pp. 2-3]

This establishes an application/shell-route association with that container. It does not independently establish that ChatGPT.exe itself performed the filesystem write or that the historical transition used the identical initiating action. Controlled association is established; historical initiation remains a separate join.

7. Process Lineage and Service Attribution

Repeated process-tree captures establish explorer.exe PID 3856 -> ChatGPT.exe PID 41488, with ChatGPT.exe PID 45036, codex.exe PID 32232, and codex-computer-use-swift.exe PID 10632 as siblings beneath the common ChatGPT root. Codex PID 32232 separately parents cmd.exe PID 43616. [Full.PDF, p. 1]

SystemSettings.exe remains in a separate svchost.exe lineage. [Full.PDF, p. 20]

DNS evidence resolved sdmntprsouthcentralus.oaiusercontent.com to 172.64.144.52 and 104.18.43.204 while Codex and ChatGPT connected to those addresses during the same reconnect episode. The defensible claim is contemporaneous hostname/IP association; without per-process DNS, TLS SNI, HTTP Host, or equivalent evidence, an individual socket-to-hostname binding is not independently established. [Full.PDF, p. 2]

A later receipt applies the same method to ab.chatgpt.com and OneDrive blob storage: process/service endpoint association is established; specific request or file payload is unknown. [Full.PDF, p. 108]

8. Cross-Surface Access

The record establishes that Codex programmatically accessed referenced ChatGPT conversation material and locally materialized attachment previews. A `codex_work_desktop` rollout invoked `read_thread` against an active ChatGPT conversation and received conversation turns; the same rollout later read an attachment from a `codex-file-preview` path. [Full.PDF, p. 92]

This converts a capability question into an execution question for those events. Documented Codex session starts also align with several dense connection bursts, making ordinary task-start activity a strong explanation for those particular bursts. [Full.PDF, p. 85] The explanation changes interpretation; it does not erase the access.

9. Proposition Substitution

The behavioral corpus documents a recurring failure mode: a correction can be accurately described and later violated. The packet characterizes this as correction not binding and records returns to procedures, safeguards, artifacts, and adjudicative framing after those moves were rejected. [Full.PDF, p. 62]

A related pattern is proposition substitution: proposition A is established, a stronger proposition B is introduced, B is declared unestablished, and the answer closes on B. The packet gives the example of an authorization inconsistency being followed by an unsolicited qualification about human intervention. [Full.PDF, pp. 61-62]

Forensic writing must resist the same error. Actor not established is relevant to actor attribution; it is not a rebuttal to an established state transition. Payload not reconstructed is relevant to payload content; it is not a rebuttal to successful file-specific upload telemetry.

10. Counter-Hypothesis Elimination

A serious forensic method must kill its own weak hypotheses. Several Codex network bursts align with documented task/session starts and requested reads, weakening their use as intrusion evidence. [Full.PDF, p. 85] Historical `Get-ProcessTag` failures were traced to investigator-created PowerShell monitoring behavior and excluded from hostile-attribution evidence. [Full.PDF, p. 1]

Fresh Open WebUI containers were reconstructed as Docker Desktop GUI workflows, and the inspected container contained no users, chats, documents, embeddings, or Windows bind mounts. That weakens the theory that that particular fresh container was used for document ingestion. [Full.PDF, pp. 88-89]

This is not defensive padding. It makes the remaining findings more durable because unsupported arrows are removed rather than protected.

11. Implications

AI-integrated desktop environments collapse boundaries that older forensic models often treated separately. A single user-visible application may include multiple executable trees, local helpers, cloud APIs, shell integrations, background synchronization, and cross-surface access to conversations or attachments.

Application-level telemetry should be preserved before it rolls over. Controlled reproduction should be used to establish application associations without pretending to reconstruct historical intent. Negative search results should be treated as coverage statements, not universal absences: the record contains an empirical counterexample in which connector Trash search returned zero while native Drive showed 24 items that were later restored. [Full.PDF, p. 57]

12. Limitations

This paper does not identify a single common external operator. It does not reconstruct every TLS payload, identify every cloud-side reader, or resolve the physical-human initiator of every historical event. Windows session evidence and provider-side account/file-access histories remain incomplete. [Full.PDF, p. 90]

Those limitations constrain attribution claims. They do not reopen whether lower-level events occurred. The record itself concludes that several genuine data-loss, authorization-consistency, synchronization, application, and network events are established while a common external actor and unified causal chain are not. [Full.PDF, p. 91]

13. Conclusion

The investigation demonstrates a general forensic principle: uncertainty should propagate forward, not backward. Missing human attribution limits an actor claim; it does not erase an exact-object destructive transition. Missing cloud-reader identity limits recipient attribution; it does not erase a successful named-file upload. Unknown hidden reviewer state may explain an authorization reversal; it does not erase the reversal.

The strongest architecture is not a single grand causal narrative. It is a set of independently established event classes connected only where the record supplies a bridge. The master packet reaches that posture directly: substantive documents became empty; automatic authorization reversed on unchanged content; monitoring evidence was synchronized through OneDrive; Codex accessed referenced conversations and attachment previews; and the ChatGPT OpenWith route was experimentally linked to the specific Jump List container under investigation. [Full.PDF, p. 103]

Forensic uncertainty should propagate into propositions that require missing evidence. It should not be allowed to erase propositions already established by lower-level receipts.

Appendix: Claim Matrix

Proposition	Status	Boundary
Substantive Drive objects later became empty	ESTABLISHED	Actor unresolved
Automatic review reversed on unchanged returned content	ESTABLISHED	Hidden decision state unresolved
Named monitoring artifacts synchronized through OneDrive	ESTABLISHED	Later reader unresolved
Remote Git object set existed while publication branch lagged	ESTABLISHED	Transfer != publication
ChatGPT OpenWith route modifies target Jump List	ESTABLISHED	Historical initiator unresolved
ChatGPT root parents Codex and helper	ESTABLISHED	Does not identify human operator
Common hidden operator caused all event classes	NOT ESTABLISHED	No complete causal/identity bridge

Source Note

Primary source: Full.PDF, 108 pages, supplied as the consolidated evidence packet. Bracketed page references refer to that packet. This manuscript intentionally states affirmative findings at the strongest level supported by the record and confines unresolved attribution to the propositions that actually require it.